

1 Forward, Futures & Hedging

Formulas

$$\sum_{i=0}^{n-1} ar^i = \frac{a(1-r^n)}{1-r}$$

For given rate $r\%$ over time period $t \in [0, T]$, with annual period P , discount factor $d_{0,T}$ is given

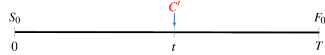
$$d_{0,T} = \left(1 + r\% \times \frac{T}{P}\right)^{-1} < 1$$

No Arbitrage Argument

To prove $A = B$, show strategies for $A \neq B$. If $A > B$, short on A , long on B , and vice versa.

1.1.2 Asset w/ Predictable Income

$$F_0 = (S_0 - C)/d_{0,T}, C = C'd_{0,t}$$

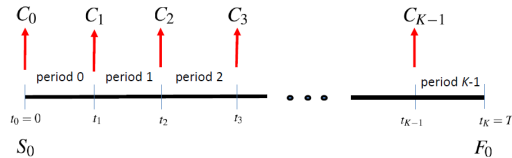


1.1.3 Asset w/ Continuous Dividend Yield

Annualized continuous dividend yield of q

$$F_0 = S_0 \exp((r - q)T)$$

1.1.4 Asset w/ Storage Costs

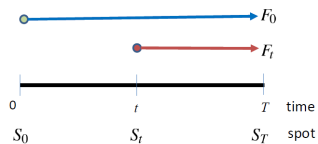


$$d_{0,K}F_0 = S_0 + \sum_{j=0}^{K-1} d_{0,j}C_j$$

1.1.5 Convenience Yield

$$d_{0,K}F_0 = S_0 + \sum_{j=0}^{K-1} d_{0,j}(C_j - y)$$

1.2 Value of Forward



Value of long position:

$$f_t = (F_t - F_0)d_{t,T}$$

1.3.1 Marked-to-Market Mechanism

- Investor deposits initial margin
- Account is marked to market each day on gain/loss of futures position
- Margin account balance fluctuates day to day
- If balance drops below threshold, investor required to top up with cash

1.3.2 Futures-Forwards Equivalence

If interest rates are known to follow expectation dynamics, then theoretical futures and forward prices of corresponding contracts are identical.

1.4.1 Short Hedge

To **lock in** a selling price, applicable when

- already owns asset and expects to sell in futures
- does not own asset but will in future, and sell later afterward

Net cash inflow of $F_0 = S_T + (F_0 - S_T)$ is locked in.

With basis risk,

$$F_1 + b_2 = S_2 + (F_1 - F_2)$$

1.4.2 Long Hedge

To **lock in** a purchase price, applicable when hedger has to purchase asset in the future.

Net cash outflow of $F_0 = S_T - (S_T - F_0)$ is locked in.

With basis risk,

$$F_1 + b_2 = S_2 - (F_2 - F_1)$$

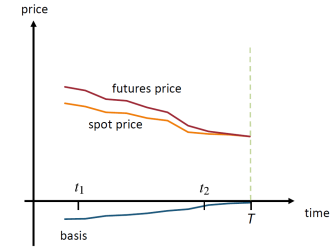
1.4.3 Basis Risk

Perfect hedges often not possible in reality.

- unavailability of futures contract matching asset obligation dates
- quantity of asset obligated may not be integer multiple of contract size
- lack of liquidity in futures market
- terms of delivery may not coincide with obligation
- unavailability of futures contracts with underlying asset exactly matching asset hedged

Define basis b_i at time t_i , with spot price S_i and futures price F_i as

$$b_i = S_i - F_i$$



Effective price can be written as

$$F_1 + b_2 = S_1 - (b_1 - b_2)$$

Price at time t_1 , offset by diff btw **initial basis** b_1 and **cover basis** b_2 .

Should choose futures contract with maturity close, but later than expiration of hedge.

1.4.4 Cross Hedging

Futures and spot price of proxy asset at time t_i denoted by F_i^* and S_i^* respectively. Effective price achieved is

$$S_2 + (F_1^* - F_2^*) = F_1^* + (S_2^* - F_2^*) + (S_2 - S_2^*)$$

1.5 Minimum Variance Hedge

At t_1 , hedger has obligation to purchase Q_a units of asset at time t_2 . Cash flow at t_2 is

$$x = Q_a(S_1 + \Delta S)$$

Suppose F is futures price of contract used as hedge, Q_f denotes size of long futures contract taken. Hedge ratio

$$h = \frac{Q_f}{Q_a}$$

At time t_2 , cash flow on hedged position

$$y = Q_a(S_1 + \Delta S - h\Delta F)$$

$$\Delta F = F_2 - F_1$$

Minimum-Variance Hedge ratio

$$h^* = \frac{\text{cov}(\Delta S, \Delta F)}{\text{var}(\Delta F)}$$

$$h^* = \rho \frac{\sigma_{\Delta S}}{\sigma_{\Delta F}}$$

$$Q_f^* = Q_a h^*$$

2 Forwards & Futures in Practice

2.1.1 US Treasury Bills

- **Dollar price:** Typically quoted per \$100 of face value
- Often given as **discount yield** d

$$d = \frac{Par - P}{Par} \times \frac{360}{D}$$

where D is days to maturity.

- Price is then

$$P = Par \times \left(1 - d \times \frac{D}{360}\right)$$

- True yield is the **coupon equivalent yield** (CEY) or **investment rate**

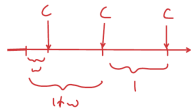
1. T-bills not more than half-year to maturity

$$i = \frac{100 - P}{P} \times \frac{y}{t}$$

2. T-bills more than half-year to maturity. Root of

$$\left(\frac{t}{2y} - 0.25\right)i^2 + \frac{t}{y}i + \frac{P - 100}{P} = 0$$

2.1.2 Accrued Interest



Bond price a.k.a. **full, dirty, or invoice price**

$$P = \frac{F}{\left(1 + \frac{r}{m}\right)^{w+n-1}} + \sum_{i=0}^{n-1} \frac{C}{\left(1 + \frac{r}{m}\right)^{w+i}}$$

Quoted prices a.k.a. **clean or flat price** does not include accrued interest.

$$DPrice = CPrice + AI$$

2.2.1 Repurchase Agreements

- Repurchase agreements = **repo** = **collateralized loan**.
- Underlying security held as **collateral** by counter-party.
- Counter-party entered **reverse repo**.
- Rate involved is **repo rate**.
- **Simple interest** rates. actual/360 in US.



1. At time $t = 0$, trader finances purchase of security worth S_0 .
2. Enter T -day repo with Mr. Y.
3. Mr. Y demands haircut of $h\%$ and repo rate r .
4. Mr. Y lends $S_0(1 - h\%)$.
5. Trader has net outflow of $S_0 \times h\%$.
6. At time $t = T$, trader pays $F = S_0(1 - h\%)(1 + r \times T/360)$.
7. Trader receives security.
8. Equivalent to forward purchase of security at forward price F .
9. If security price increases to S_T then trader earns profit.
10. To carry out bet on decline using reverse repo, buy at S_0 , sell at $S_0(1 - h\%)$, buy back at S_T .

2.2.2 SOFR

Secured Overnight Financing Rate (SOFR). Day count: actual/360. Compound SOFR Formula

$$r = \left[\prod_{i=1}^T \left(1 + \frac{r_i n_i}{Y}\right) - 1 \right] \frac{Y}{d_c}$$

Simple SOFR Formula

$$r = \frac{1}{d_c} \sum_{i=1}^T r_i n_i$$

2.2.3 1-Month and 3-Month SOFR Futures

- 1-month SOFR Futures: \$5m loan for 1-month
- 3-month SOFR Futures: \$1m loan for 3-months
- SOFR Futures follow 30 day periods
- Quoted Futures price 1-month: $100 \times (1 - E(r_m))$, where r_m is the simple average SOFR for contract month
- Final Settlement price 1-month: $100 \times (1 - r_m)$.
- Quoted Futures price 3-months: $100 \times (1 - \text{expected compound average SOFR for contract period})$
- Long position gains when expected SOFR falls (futures rises).
- Take up long position to hedge against rate falls.
- Long position is equivalent to agreeing to lend at rate

$$(100 - F_0)\%$$

- Short position gains when expected SOFR rises (futures falls).
- Take up Short position to hedge against rate rises.
- Short position is equivalent to agreeing to borrow at rate

$$(100 - F_0)\%$$

Changes to margin account

- 1-month: Decrease in futures price by 0.01 (rate rise by 1 bp) results in long position being decreased by \$41.67.
- 3-months: Decrease in futures price by 0.01 (rate rise by 1 bp) results in long position being decreased by \$25.

